c-only slope law: the output matrix H

1 From the state-space model to $\mathbf{F}_e$ and $\mathbf{H}$

Linear

$$\mathbf{x}[k+1] = \mathbf{A}\,\mathbf{x}[k] + \mathbf{B}\,u[k] + \mathbf{w}[k] \qquad \hat{\mathbf{x}}[k+1] = \mathbf{A}\,\hat{\mathbf{x}}[k] + \mathbf{B}\,u[k]$$
$$\mathbf{y}[k] = \mathbf{C}\,\mathbf{x}[k] + \mathbf{D}\,u[k] + \mathbf{v}[k] \qquad \hat{\mathbf{y}}[k] = \mathbf{C}\,\hat{\mathbf{x}}[k] + \mathbf{D}\,u[k]$$

$$\begin{aligned}\mathbf{e}[k+1] &= \mathbf{A}\,\mathbf{e}[k] + \mathbf{w}[k] \\ \mathbf{e}_y[k] &= \mathbf{C}\,\mathbf{e}[k] + \mathbf{v}[k]\end{aligned} \qquad \Longrightarrow \qquad \mathbf{F}_e = \mathbf{A}, \qquad \mathbf{H} = \mathbf{C}$$

Nonlinear

$$\mathbf{x}[k+1] = \mathbf{f}\big(\mathbf{x}[k],\, u[k]\big) + \mathbf{w}[k] \qquad \hat{\mathbf{x}}[k+1] = \mathbf{f}\big(\hat{\mathbf{x}}[k],\, u[k]\big)$$
$$\mathbf{y}[k] = \mathbf{g}\big(\mathbf{x}[k],\, u[k]\big) + \mathbf{v}[k] \qquad \hat{\mathbf{y}}[k] = \mathbf{g}\big(\hat{\mathbf{x}}[k],\, u[k]\big)$$

$$\begin{aligned}\mathbf{e}[k+1] &\approx \left.\frac{\partial \mathbf{f}}{\partial \mathbf{x}}\right|_{\hat{\mathbf{x}}[k],\,u[k]} \mathbf{e}[k] + \mathbf{w}[k] \\ \mathbf{e}_y[k] &\approx \left.\frac{\partial \mathbf{g}}{\partial \mathbf{x}}\right|_{\hat{\mathbf{x}}[k],\,u[k]} \mathbf{e}[k] + \mathbf{v}[k]\end{aligned} \qquad \Longrightarrow \qquad \mathbf{F}_e[k] = \left.\frac{\partial \mathbf{f}}{\partial \mathbf{x}}\right|_{\hat{\mathbf{x}}[k],\,u[k]}, \qquad \mathbf{H}[k] = \left.\frac{\partial \mathbf{g}}{\partial \mathbf{x}}\right|_{\hat{\mathbf{x}}[k],\,u[k]}$$

c-only: input $u[k] = \Delta\bar{w}_d[k]$, $\nabla_d \bar{w}_d[k] = \sum_{i=0}^{d-1} \Delta\bar{w}_d[k-i]$

$$f_4(\mathbf{x}, u) = \bar{a}_w + \frac{1-c}{h^2}\,M, \qquad M = \Delta\bar{w}_d + (1-\lambda_c)\,\delta\bar{w}_3$$
$$g_2(\mathbf{x}, u) = \bar{a}_w - \frac{1-c}{h^2}\,\nabla_d \bar{w}_d, \qquad h = \bar{w}_d - \delta\bar{w}_3$$

2 State and output

$$\mathbf{x}[k] = \begin{bmatrix} \delta\bar{w}_1[k] \\ \delta\bar{w}_2[k] \\ \delta\bar{w}_3[k] \\ \bar{a}_w[k] \\ c[k] \end{bmatrix}, \qquad \begin{aligned} y_1[k] &= \delta\bar{w}_1[k] + n_w[k] \\ y_2[k] &= \bar{a}_w[k-d] + n_a[k] \end{aligned}$$

3 y_2 as a function of the current state

$$\bar{a}_w[k-d] \approx \bar{a}_w[k] - \bar{a}'\big(h[k];\, c[k]\big)\,\nabla_d \bar{w}_d[k]$$
$$\bar{a}'(h;\, c) = \frac{1-c}{h^2}$$
$$h[k] = \bar{w}_d[k] - \delta\bar{w}_3[k]$$
$$\nabla_d \bar{w}_d[k] = \bar{w}_d[k] - \bar{w}_d[k-d]$$

$$y_2[k] = g\big(\mathbf{x}[k]\big) + n_a[k], \qquad g(\mathbf{x}) = \bar{a}_w - \frac{1-c}{\big(\bar{w}_d - \delta\bar{w}_3\big)^2}\,\nabla_d\bar{w}_d$$

4 Predicted output and output error

$$\hat{y}_1[k] = \delta\hat{\bar{w}}_1[k]$$
$$\hat{y}_2[k] = g\big(\hat{\mathbf{x}}[k]\big) = \hat{\bar{a}}_w[k] - \frac{1-\hat{c}[k]}{\hat{h}[k]^2}\,\nabla_d\bar{w}_d[k], \qquad \hat{h}[k] = \bar{w}_d[k] - \delta\hat{\bar{w}}_3[k]$$

$$e_{y_1}[k] = y_1[k] - \hat{y}_1[k] = e_{\delta\bar{w}_1}[k] + n_w[k]$$
$$e_{y_2}[k] = y_2[k] - \hat{y}_2[k] = g\big(\mathbf{x}[k]\big) - g\big(\hat{\mathbf{x}}[k]\big) + n_a[k]$$

5 First-order Taylor of g at the estimate

$$g(\mathbf{x}) - g(\hat{\mathbf{x}}) \approx \sum_{j=1}^{5} \frac{\partial g}{\partial x_j}\bigg|_{\hat{\mathbf{x}}} \big(x_j - \hat{x}_j\big) = \frac{\partial g}{\partial \mathbf{x}}\bigg|_{\hat{\mathbf{x}}} \mathbf{e}$$
$$\mathbf{e} = \mathbf{x} - \hat{\mathbf{x}} = \big[\, e_{\delta\bar{w}_1},\ e_{\delta\bar{w}_2},\ e_{\delta\bar{w}_3},\ e_{\bar{a}_w},\ e_c \,\big]^T$$

6 The partial derivatives

$$\frac{\partial g}{\partial\, \delta\bar{w}_1} = 0$$
$$\frac{\partial g}{\partial\, \delta\bar{w}_2} = 0$$
$$\frac{\partial g}{\partial\, \delta\bar{w}_3} = -\nabla_d\bar{w}_d\,\frac{\partial\bar{a}'}{\partial h}\,\frac{\partial h}{\partial\, \delta\bar{w}_3} = -\nabla_d\bar{w}_d\Big(-\frac{2(1-c)}{h^3}\Big)(-1) = -\frac{2\bar{a}'}{h}\,\nabla_d\bar{w}_d$$
$$\frac{\partial g}{\partial \bar{a}_w} = 1$$
$$\frac{\partial g}{\partial c} = -\nabla_d\bar{w}_d\,\frac{\partial\bar{a}'}{\partial c} = -\nabla_d\bar{w}_d\Big(-\frac{1}{h^2}\Big) = \frac{\nabla_d\bar{w}_d}{h^2}$$

7 Output matrix

$$\mathbf{H}[k] = \begin{bmatrix} \dfrac{\partial\hat{y}_1}{\partial\mathbf{x}} \\[2ex] \dfrac{\partial g}{\partial\mathbf{x}}\Big|_{\hat{\mathbf{x}}[k]} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\dfrac{2\bar{a}'[k]}{\hat{h}[k]}\,\nabla_d\bar{w}_d[k] & 1 & \dfrac{\nabla_d\bar{w}_d[k]}{\hat{h}[k]^2} \end{bmatrix}, \qquad \bar{a}'[k] = \frac{1-\hat{c}[k]}{\hat{h}[k]^2}$$

$$\begin{bmatrix} e_{y_1}[k] \\ e_{y_2}[k] \end{bmatrix} = \mathbf{H}[k]\,\mathbf{e}[k] + \begin{bmatrix} n_w[k] \\ n_a[k] \end{bmatrix}$$

Hold ($\nabla_d \bar{w}_d = 0$)

$$\mathbf{H}[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

8 As implemented (y_2 whitened, echo factor)

$$y_2[k] = a_{cov}\, u[k] \qquad \text{(whitened reading, } s = a_{cov})$$

$$\hat{y}_2[k] = s\Big(\hat{\bar{a}}_w[k] - (1 - S[k])\, \bar{a}'[k]\, \nabla_d \bar{w}_d[k]\Big)$$

$$\mathbf{H}_2[k] = s\,(1 - S[k])\, \Big[\, 0,\ 0,\ -\frac{2\bar{a}'[k]}{\hat{h}[k]}\, \nabla_d \bar{w}_d[k],\ 1,\ \frac{\nabla_d \bar{w}_d[k]}{\hat{h}[k]^2}\, \Big]$$

$$\frac{\partial \hat{y}_2}{\partial \bar{a}_w} - H_{24} = s\, S[k]$$